NITRO COMPOUNDS

Chemical properties of Nitro alkane (RNO_2)

Reduction

$R - NO_2$ → (Sn / HCl, acid) → RNH_2

$R - NO_2$ → (Zn / NH_4Cl, Neutral) → $RNHOH$

$R - NO_2$ → (Catalyst) → $R - NH_2$

Hydrolysis

$RCH_2NO_2 \xrightarrow[\Delta]{HCl / H_2O} RCOOH$

$R_2CHNO_2 \xrightarrow{HCl / H_2O} R_2CO + N_2O + H_2O$ (Ketone)

Halogenation

$CH_3NO_2 \xrightarrow[NaOH]{Cl_2} CCl_3NO_2 + 3\,HCl$

Chloropicrin

Methods of preparation of $R - NO_2$ and RONO

From alkyl halides

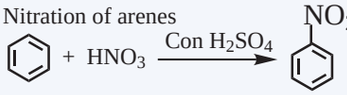
$R - X \xrightarrow{2\,AgNO_2} RNO_2 + RONO + 2AgI$ (Nitroalkane (Major), Alkylnitrite (Minor))

$R - X \xrightarrow{KNO_2} RNO_2 + R - O - N = O$ (Minor) (Major)

Nitration of alkane

$CH_3 - CH_3 \xrightarrow{HNO_3} CH_3CH_2NO_2 + CH_3NO_2$

Nitration of arenes



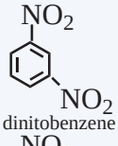
$+ HNO_3 \xrightarrow{Con\ H_2SO_4}$

$HNO_3 + H_2SO_4$ nitratingmixture

NO_2^+ electrophile

Electrophilic Substituion Reaction

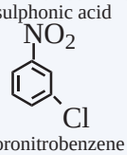
Nitrogroup is meta - directing

Nitration: $Conc.HNO_3 + H_2SO_4$ → 

m - dinitobenzene

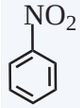
Sulphonation: $Conc.H_2SO_4$ →

m - nitrobenzene sulphonic acid

Halogenation: Cl_2 / Fe → 

m - chloronitrobenzene

Nitrobenzene (oil of mirabane) reduction



Acidmedium, Sn / HCl → $C_6H_5NH_2 + H_2O$ Aniline

Neutral, $Zn + NH_4Cl$ → C_6H_5NHOH N - Phenylhydroxylamine

Alkaline medium, $Na_3AsO_3/NaOH$ → $C_6N_5 - N = N - C_6H_5$ Azoxybenzene

Zn / NaOH / CH_3OH (or) $LiA1H_4$ → $C_6H_5 - N = N - C_6H_5$ Azobenzene

Zn / NaOH → $C_6H_5 - NH - NH\,C_6H_5$ Hydrazobenzene

Electrolytic reduction: Weaklyacidic, Medium → $C_6H_5NH_2$ Aniline

Stronglyacidic, $Conc\ H_2SO_4$ → $H_2N - C_6H_4 - OH$ P - nitrophenol

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